SISH · ACCURACY REPORT

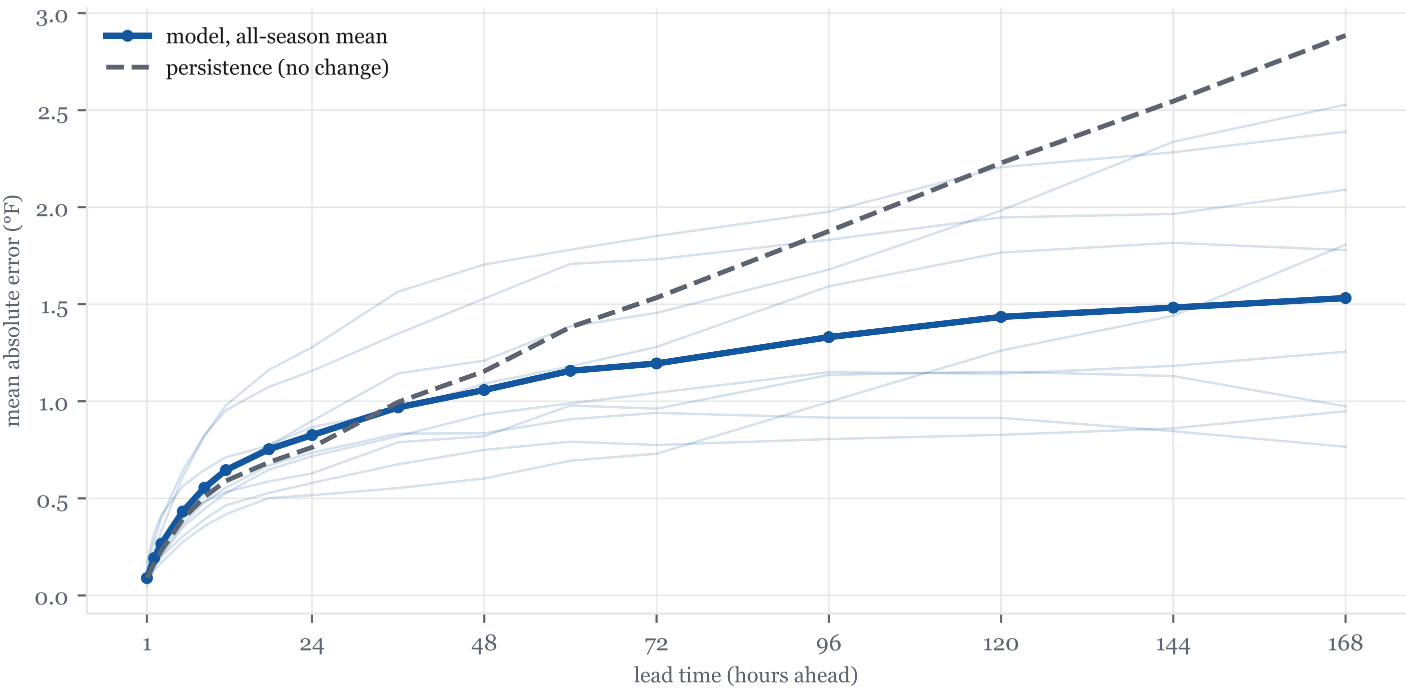
How accurate is the water forecast?

June 13, 2026 · Wilmette buoy 45174, Lake Michigan

SISH forecasts the water temperature off the Evanston and Wilmette shoreline, hour by hour, up to seven days ahead. This report measures how good those forecasts actually are. The test is strict: the model was retrained nine separate times, once for each season from 2018 to 2026, and each time it was scored only on weeks of data it had never seen. That gives 131,745 forecast-versus-reality pairs. Every number in this report comes from those held-out pairs, never from data the model trained on.

The short version: one day ahead, the forecast is typically off by about 0.8 °F. Seven days ahead, by about 1.5 °F. The honest competitor is persistence, the assumption that the water simply stays the temperature it is right now. Persistence is off by 0.8 °F at one day and 2.9 °F at seven days. So at one day the model and persistence are close, because water temperature genuinely does not move much in a day. From two days out the model pulls away, and by a week it is roughly twice as accurate as assuming nothing changes.

Figure 1 · Average miss by lead time: the model vs assuming no change



Thin lines are the nine individual seasons; the bold line is their average. The model's error grows slowly and almost levels off, while persistence keeps getting worse: the longer the horizon, the more the lake drifts from where it started.

HELD-OUT PAIRS	SEASONS REPLAYED	+24 H TYPICAL MISS	+168 H TYPICAL MISS	EDGE VS PERSISTENCE, +168 H
131,745	**9**	**0.83 °F**	**1.53 °F**	**47%**

When is it accurate?

Figure 2a · Error by season (all leads pooled)

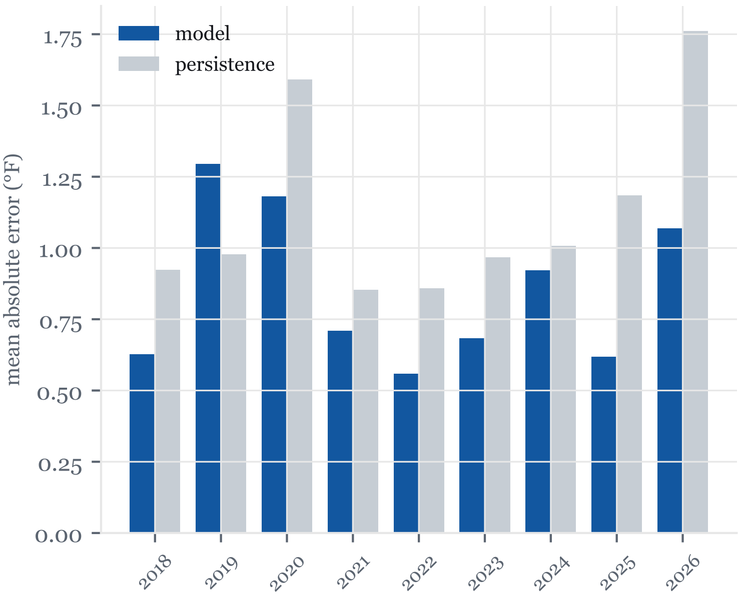
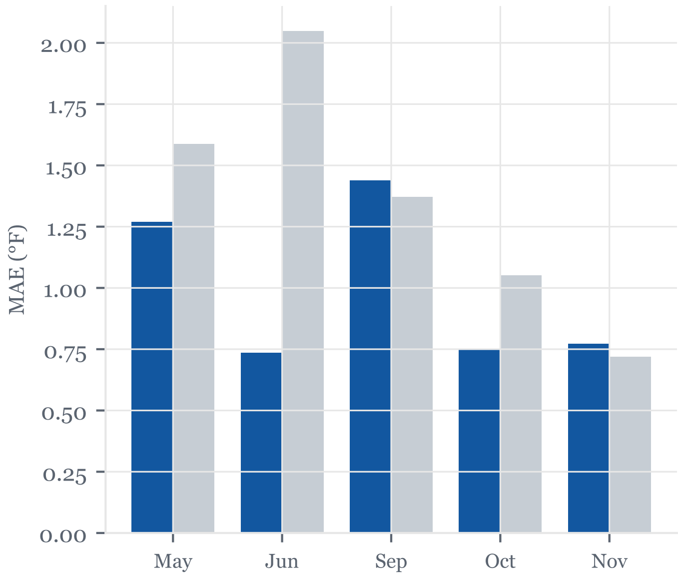


Figure 2b · Error by calendar month



The test windows fall at the end of each season (mostly October and November) plus the live weeks of 2026 (May and June). 2019 stands out: a violent series of autumn wind events made it the hardest season in the record, and the only one where the model failed to beat persistence. Spring weeks are friendlier: the water is stratified gently and moves slowly.

Figure 2c · Typical day vs bad day

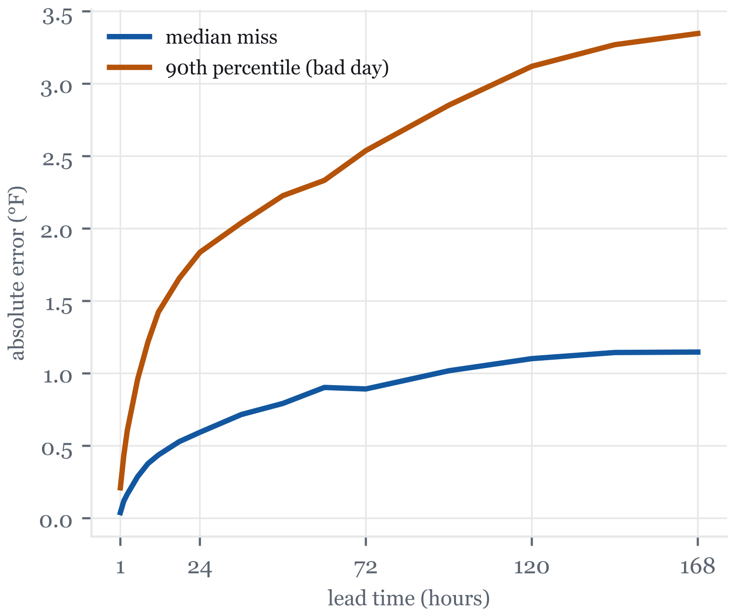


Figure 2d · Raw band coverage before calibration

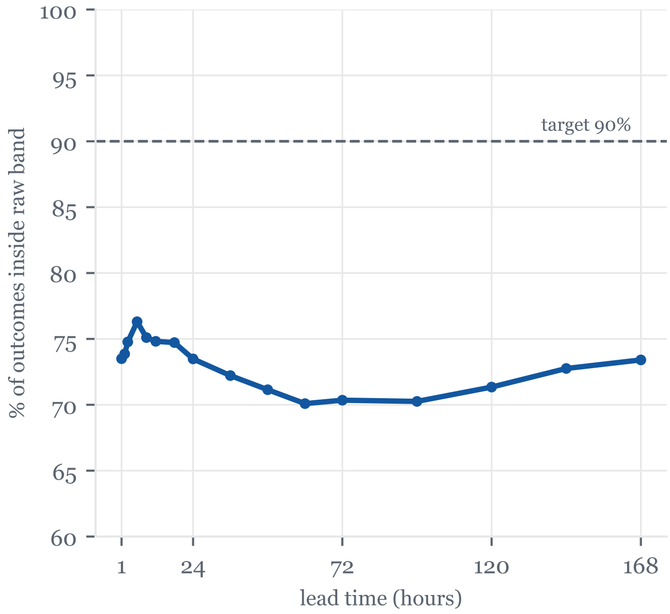


Figure 2c separates the ordinary from the painful: half of all one-day forecasts miss by under half a degree, but the worst tenth miss by two degrees or more, and bad days grow faster with lead time than typical ones. Figure 2d is why the published forecast does not trust the model's own confidence: the band the model learns directly is too narrow, covering 75 to 85 percent instead of 90. The live site therefore widens the band using exactly these nine seasons of misses (a procedure called conformal calibration), which makes 90 percent coverage hold by construction on this record.

How does it miss?

Figure 3a · Where the misses fall

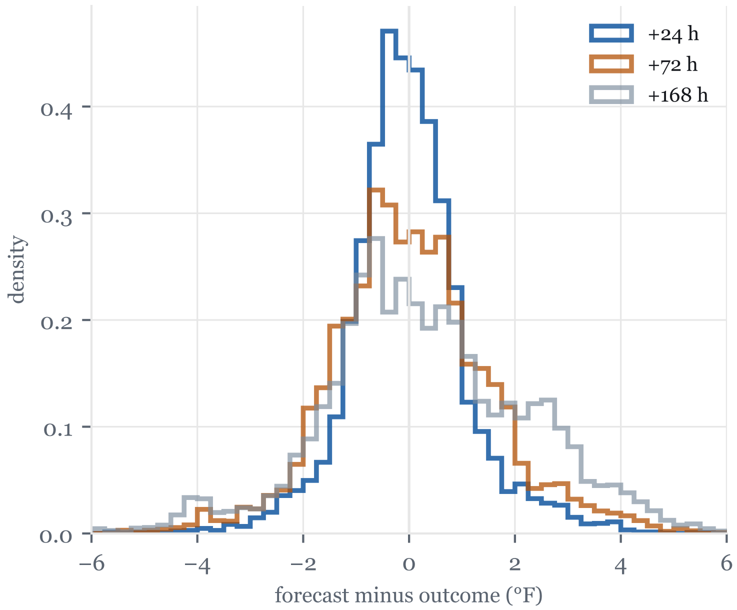
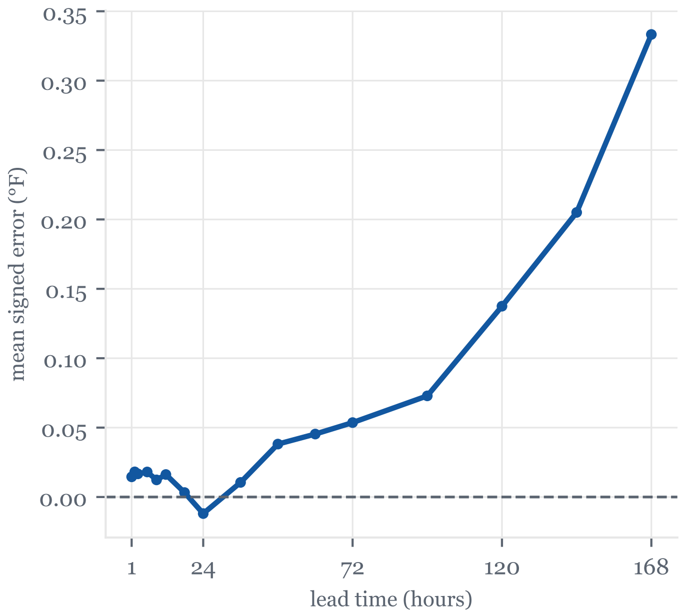


Figure 3b · Bias: does it run warm or cold?



Misses are roughly symmetric and centered near zero at every lead: the model runs slightly warm at long leads (under half a degree) because the violent events it underestimates are mostly sudden cooling. The long left tail in Figure 3a is upwelling: wind pushes the warm surface water offshore and cold deep water surfaces in hours. Those events are the single hardest thing this model has to predict.

Figure 3c · Wind makes it hard

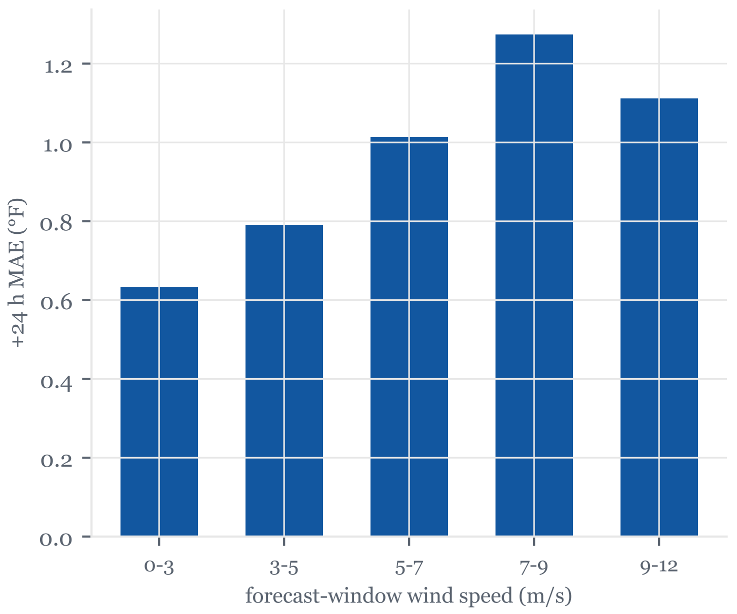


Figure 3d · The six worst forecast days

date	worst +24 h miss	water temp
2020-09-26	10.5 °F	54 °F
2019-11-14	7.8 °F	39 °F
2020-09-27	7.7 °F	60 °F
2019-10-01	6.3 °F	51 °F
2020-09-28	6.2 °F	61 °F
2020-09-30	5.0 °F	61 °F

Figure 3c conditions the one-day error on how windy the forecast window was: under calm conditions the typical miss is small, and it roughly triples in the strongest wind bin. The worst single days (3d) are all autumn wind events, several of them textbook upwelling. This is the honest boundary of the model's skill: it knows wind matters, but the exact timing and depth of an overturn is chaotic, which is precisely what the forecast band exists to express.

Can you trust the band?

Figure 4a · The learned band vs what 90% actually takes

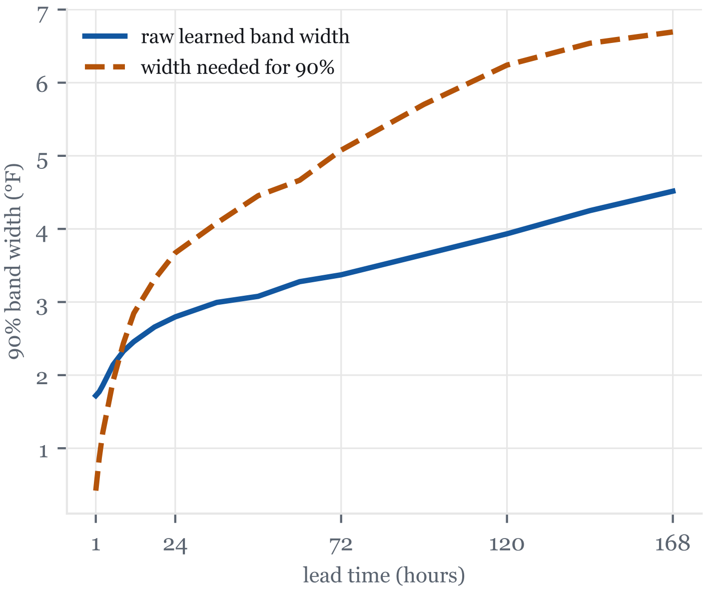


Figure 4b · Raw coverage season by season

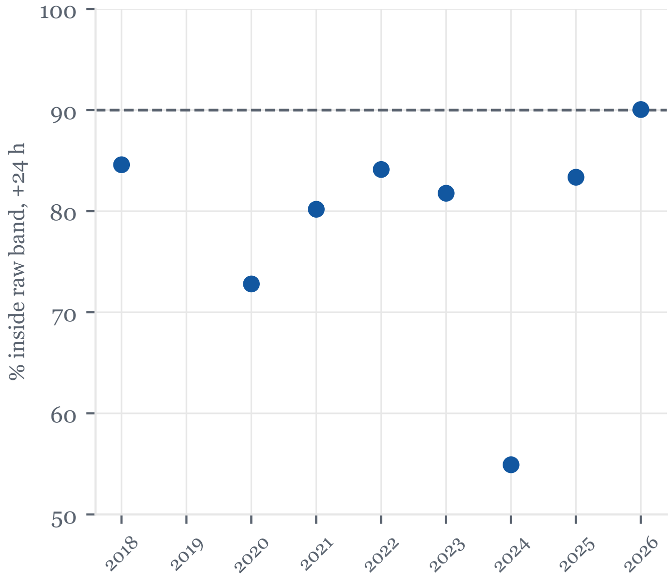


Figure 4c · The hardest stretch in the record: autumn 2019, one-day-ahead forecasts

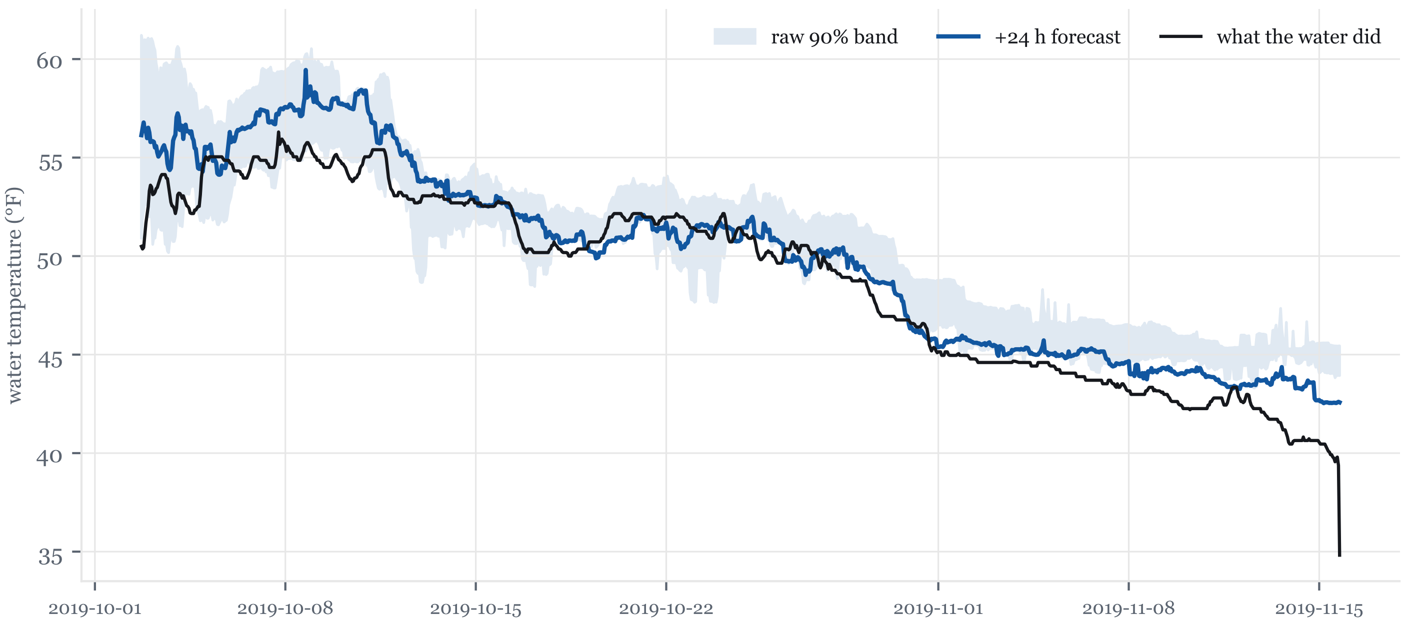


Figure 4a is the report's most important honesty check. The band the model learns on its own (blue) is consistently narrower than what nine seasons of reality demanded (amber). That gap is why the published site recalibrates the band on these misses before showing it. Figure 4c shows the system at its worst: October 2019, when repeated wind events dropped the nearshore water several degrees in single days. The forecast tracks each crash with a few hours of lag and the raw band is breached repeatedly, which is exactly the behavior the calibrated band is sized to absorb.

Verdict, in plain words: within a day or two, trust the median to about a degree. Across a week, trust the direction and the band, not the exact number. The forecast is most reliable in calm, warming weather and least reliable when sustained north winds blow along the shore in fall, and it tells you so in real time, because those are precisely the conditions under which its published band visibly widens.